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Roll No

EC-7005(1)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Information Theory and Coding

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Define uncertainty, information and entropy. State the various units of information and find relation between them. Show that entropy of a binary system (two events) is maximum when both events are equiprobable.
- b) Consider that two sources S_1 and S_2 emit message x_1, x_2, x_3 and y_1, y_2, y_3 with the joint probability $p(X, Y)$ as shown in the matrix form. Calculate $H(X)$, $H(Y)$, $H(X/Y)$ and $H(Y/X)$. Given that:

		y_1	y_2	y_3
$p(Y/X) =$	x_1	3/40	1/40	1/40
	x_2	1/20	3/20	1/20
	x_3	1/8	1/8	3

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2. a) Apply the Huffman encoding procedure for the following message ensembles:

$$[X] = [X_1 \quad X_2 \quad X_3 \quad X_4 \quad X_5 \quad X_6 \quad X_7]$$

$$[P] = [0.4 \quad 0.04 \quad 0.08 \quad 0.12 \quad 0.08 \quad 0.08 \quad 0.2]$$

Find the codewords for every message and calculate the efficiency of the coding method. The number of symbol used for coding $M = 3$.

- b) Explain the Shannon-Fano coding method with the help of a suitable example.
3. a) Explain syndrome and syndrome decoding method in linear block codes with suitable diagram.
- b) A parity check matrix is given as

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

- i) Determine generator matrix
- ii) Find code word that begins with 011
- iii) Suppose received vector is 110110, decode the vector
4. a) What are cyclic codes? Write down the basic properties of cyclic codes.
- b) The generator polynomial for (7, 4) cyclic Hamming code is given by $G(x) = 1 + x + x^3$. Determine the systematic and non systematic code word for the message vector 1010.
5. The generator polynomial of a (7, 4) cyclic code is $g(x) = 1 + x + x^2$. Find the code-words of this code.

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6. Explain Viterbi algorithm for maximum likelihood decoding.
7. Given that Message (D) = 1010001101 and Pattern (P) = 110101, calculate the FCS (R) and the message which is actually transmitted (T) using Cyclic Redundancy Check method. Also check it for correctness at receiver end that received message is correct or not. Do all calculations in clear manner.
8. Write short notes on:
 - a) Hamming code and its applications
 - b) Convolutional codes
